

Factor Tree Relay

Unit 1: Number System Launch · Lesson 1-1 · EduWonderLab

UNIT 1 · LESSON 1-1 · TEACHER NOTES

Standard 6.NS.4

FORMAT Partner relay / card game

TIME 25–30 min

GROUPING Pairs

TOPIC Prime Factorization

STANDARD 6.NS.4

PREP LEVEL Light — print & cut number cards

Learning goal *I can write the prime factorization of a number using a factor tree.*

Teacher setup

- Print one card set and one recording sheet per pair. Cut the number cards apart and shuffle.
- Each partner needs a different color pencil so you can tell whose tree is whose.
- Decide who is Builder and who is Checker for round 1, then switch each round.

Materials

Number cards (cut apart), Factor-tree recording sheet, Colored pencils, Pencil.

Differentiation & language support

- Sentence stem for every card: “I know ___ is prime/composite because ___.”
- Offer a small multiplication chart to students who are still building fact fluency.
- Pre-teach prime vs. composite with 2, 3, 4, 6 before starting (SPED/ESOL).

Key vocabulary (English / Español):

Term	Español	What it means
prime	primo	a number with exactly two factors: 1 and itself
composite	compuesto	a number with more than two factors
factor	factor	a number you multiply
prime factorization	factorización prima	a number written as a product of only primes

Quick teacher check: *Scan one factor tree per pair. Watch for branches that stop at a composite number (like 4 or 9).*

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Name: _____ Date: _____ Class: _____

Your goal: *I can write the prime factorization of a number using a factor tree.*

What you need

Number cards, Factor-tree recording sheet, Colored pencils.

How to play

1. Builder draws a number card and builds a factor tree until every branch ends in a prime.
2. Write the prime factorization at the bottom using exponents (for example, $2^3 \times 3$).
3. Checker rebuilds the same number a different way to prove the prime factorization matches.
4. If you agree, the Builder earns a “prime stamp” on the recording sheet. Then switch roles.
5. Keep playing until all cards are stamped.

Number Cards

Draw a card, build the factor tree, then write the prime factorization using exponents.

1. 36	2. 48
3. 60	4. 72
5. 90	6. 84
7. 100	8. 54
9. 96	10. 120

Record your work (one factor tree per card):

Explain your thinking

Pick one card. Explain how you knew when to stop building your factor tree.

Sentence frame: *I stopped building when every branch ended in a ___, because a prime number only has ___ factors.*

★ **Challenge:** Build two different-looking factor trees for 72. Show that both give the same prime factorization.

ANSWER KEY

Teacher copy — please do not distribute to students.

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Answers

1. 36 Answer: $2^2 \times 3^2$	2. 48 Answer: $2^4 \times 3$
3. 60 Answer: $2^2 \times 3 \times 5$	4. 72 Answer: $2^3 \times 3^2$
5. 90 Answer: $2 \times 3^2 \times 5$	6. 84 Answer: $2^2 \times 3 \times 7$
7. 100 Answer: $2^2 \times 5^2$	8. 54 Answer: 2×3^3
9. 96 Answer: $2^5 \times 3$	10. 120 Answer: $2^3 \times 3 \times 5$

Teaching notes & look-fors

- Order of factors does not matter, but the prime factorization is unique.
- Common error: stopping at a composite (e.g., 4) instead of breaking it into 2×2 .